

Bhagyesh Rathi

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EDUCATION

San Jose State University

Master of Science in Artificial Intelligence

San Jose, CA

Expected: May 2027

Relevant coursework: Machine Learning, Deep Learning, AI and Data Engineering, Natural Language Processing, AI threat Intelligence

San Jose State University

Bachelor of Science in Computer Science, Magna Cum Laude

San Jose, CA

August 2020 - May 2024

Relevant coursework: Java OOP, Data Structures and Algorithms, Object-Oriented Design, Software Engineering, Operating Systems, Relational Databases(SQL), Machine Learning, Advanced Programming in Python, Data Visualization, Information Security.

RELEVANT EXPERIENCE

Rakuten - Software Engineer Intern | San Mateo, CA

August 2023 – December 2023

- Developed **event listeners** in **Kotlin** to handle consent-revoked and delete-account events from Google, Facebook and Apple.
- Implemented features to validate and decode **JWTs** to extract required data, and deserialize **JSON** responses.
- Built an **SDK** to abstract low level service to service calls, published artifact in **Artifactory** through **GitLab CI/CD**.
- Achieved **85%** test coverage(**JaCoCo**) with **Unit**, **Integration testing**, and leveraged **SonarQube** to eliminate code smells.
- Deployed **microservices** on **Google Cloud Platform (GCP)** using **Gitlab CI/CD** automated build, test, and deploy pipelines.
- Created **Dockerfiles** and orchestrated containers using **Kubernetes**, deploying across non-prod and prod environments.
- Integrated **OpenTelemetry** and **Google cloud Trace** for real time **monitoring** and performance analysis.

Rakuten - Software Engineer Intern | San Mateo, CA

May 2023 – August 2023

- Contributed to **Social Sign Up / Login** with Google, Apple, Facebook for Rakuten businesses in Spain, UK, and Germany.
- Achieved a **40%** adoption rate of Social Authentication and a **37%** increase in CVR over Username and Password Authentication.
- Developed **Microservice** in **Kotlin** for social link(with Google, Facebook, Apple) management (Display and Unlink) features.
- Implemented **REST APIs** leveraging **OpenID Connect & OAuth2.0** to implement token based **Authentication and Authorization**.
- Implemented structured logging using **Log4j**, analyzed the logs through **GCP Cloud Monitoring**.
- Participated in **Agile software development** such as daily stand-ups, sprint planning, and sprint review.

San Jose State University - Teaching Assistant, CS department | San Jose, CA August 2022 – December 2022

- Conducted a 3-hour weekly lab to help **50+** students apply **OOP** concepts in **Java** and develop object-oriented thinking.
- Conducted weekly office hours to solve homework or lab-related problems and helped students over email.
- Worked closely with CS department staff and multiple professors to design and improve the lab.

PROJECTS

RAG-Powered Interactive Portfolio | Next.js, Pinecone, Vertex AI, Vercel AI SDK, Langchain | [GitHub](#), [Chat](#)

Feb 2026

- Shipped a production-grade **RAG** chatbot serving real users on a live portfolio site with full ingestion pipeline.
- PDF ingestion → recursive chunking → 768-dim embeddings → Pinecone upsert, with top-K retrieval, score thresholding, and metadata filtering to minimize hallucination
- Built a streaming API endpoint using **Vercel AI SDK** with **Gemini 2.5 Flash**, achieving sub-second **TTFT**.
- Implemented multi-layer security: input validation, system prompt hardening against **prompt injection**, and topic-scoped guardrails
- Authored a **27-test integration suite** covering **embedding quality**, **retrieval accuracy**, **pipeline correctness**, and **injection defense**.

SRE Copilot: AI Incident Triage Agent for On-Call SREs | Python, PyTorch, LangChain, ChromaDB | [Colab](#)

Feb 2025

- Built a **ReAct tool-use agent** that triages production incidents across 5 data sources (**RAG** over 768 runbooks, text-to-SQL, GCP status API, vulnerability DB, web search) using dual **Qwen 2.5** models (32B + 3B) on a single GPU via **4-bit quantization**.
- Evaluated **4 prompting techniques across 80 benchmarks**, finding self-reflection improved answer **grounding** by **125%** (0.40→0.90) and **actionability** by **100%** (0.50→1.0) over baseline, while prompt chaining achieved perfect tool selection on the 32B model but failed on 3B — revealing minimum capability thresholds for structured data.
- Achieved **~4%** prefill speedup on the **32B** model by building **KV cache** reuse for **system prompt acceleration**.
- Implemented **defense-in-depth security** (SQL injection blocking, prompt hardening) passing 3/5 red-team attacks per model; deployed via Gradio with real-time model/technique comparison.

AutoML Studio: Full-Stack Platform for Classification & Regression with Code Generation | Python, Flask, scikit-learn, XGBoost, React 18 | [GitHub](#)

Dec 2024

- Built a dual-mode AutoML platform that auto-detects classification vs regression and trains 6 classification models (LogReg, RF, SVM, KNN, XGBoost, DT) and 5 regression models in parallel via Flask background threads, exporting the winner as both a serialized .pkl and a standalone reproducible Python script — eliminating vendor lock-in vs Google AutoML / H2O.
- Engineered a hardware-aware leaderboard by spawning a psutil background thread sampling RAM/CPU every 200 ms during each training run, enabling sort-by-memory-footprint alongside accuracy and training time — a production-budget signal that accuracy-only AutoML tools ignore.
- Optimized for scale with adaptive RandomizedSearchCV iteration budgets that grow with parameter-grid size, automatic SVM→SGD substitution on >2K-row datasets to avoid $O(n^2)$ training blowup, stratified sampling that preserves class ratios on downsampled data, and a chunk-based CSV reader handling 1 GB+ files without OOM.
- Implemented imbalance-aware training (auto class_weight='balanced' for sklearn, dynamic scale_pos_weight for XGBoost) and deployed a React 18 frontend with worker-based PapaParse for non-blocking uploads, Recharts ROC / confusion-matrix / regression-fit visualizations, and a glassmorphism UI streaming live training logs from the backend.

Caption Lens: Visual-Semantic Alignment via Attention Mechanisms | PyTorch, Hugging Face | [App / Repo](#)

May 2025

- Designed and trained three **encoder-decoder image captioning** models (Show-and-Tell, Show-Attend-and-Tell with soft attention, Visual Sentinel adaptive attention) on **MS COCO 2014 (82K images, ~9K-word vocab)** using a fine-tuned **ResNet-101** encoder feeding $7 \times 7 \times 2048$ spatial features into an **LSTM** decoder.
- Implemented Bahdanau soft attention with doubly stochastic regularization, boosting **CIDEr** from **1.023** → **1.040** and winning all 8 metrics (**BLEU-1-4, METEOR, ROUGE-L, CIDEr, SPICE**); **BLEU-4** of 0.329 exceeds the original 2015 paper (0.250).
- Engineered beam search with 1–10 beam-width ablation lifting CIDEr from 1.013 (greedy) to 1.093 (beam=6); built interpretable attention heatmaps tracing each word to its driving image region.
- Deployed as an interactive multi-model demo on **Hugging Face Spaces** via **Docker** + **Gradio** with real-time model switching.

Skin Cancer Detection using CNN | Python, Tensorflow, Scikit-Learn, Pandas, Seaborn | [Colab](#)

June 2024

- Categorized 10,015 skin lesion images into 7 cancer classes, visualizing data with **Seaborn** for better understanding.
- Preprocessed data through, image resizing, **normalization, one-hot encoding, data augmentation**.
- Performed minority class **oversampling** to address class imbalance.
- Developed and implemented a **Deep Convolutional Neural Network (CNN)** with multiple layers.
- Conducted **hyperparameter tuning**, created **train-test-validation splits**, and employed **Adam optimizer** and learning rate annealer to optimize model performance, achieving **80% accuracy**.

Bank Churn Data Analysis and Prediction using ML | *Pandas, Seaborn, Matplotlib, Scikit-Learn* | [Colab](#)

Feb 2024

- Analyzed data of 10,000 account holders at a Multinational Bank by doing exploratory data analysis with **Pandas**.
- Visualized data using **Seaborn** and **Matplotlib** to choose strategies for developing ML training pipelines.
- Implemented data **preprocessing, feature selection**, established test, and training sets, and constructed a streamlined pipeline for training 5 machine learning models to predict customer churn using **Scikit-Learn**.
- Implemented **XGBoost, Random Forest, KNN, SVM, and Naive Bayes Machine Learning** models.
- Utilized **N-fold cross-validation, F1-score, confusion metrics** to evaluate the performance of each model.
- Tested the ML models using a test set to show the performance of the models and compare them to get the most accurate model.

SKILLS & CERTIFICATIONS

Programming Languages: Java, Python, JavaScript, SQL

Frameworks: Flask, Scikit-Learn, Tensorflow, Pytorch, Pandas, Numpy, Seaborn

AI/ML Tools & Frameworks: Vertex AI, Google ADK, HuggingFace, LangChain, Ollama, Claude code, Gemini, Copilot, Cursor

Skills: Distributed Systems, Multi-tiered Systems, Agile software development, Unit/Integration Testing, Monitoring and Alerting.

Databases: MongoDB, MySQL, PostgreSQL, Redis

Cloud: GCP, Docker, Kubernetes, AWS, Vercel

Certifications: [Machine Learning Specialization\(Stanford\)](#)